The Edison Institute research compendium: eight pillars of evidence

The Edison Institute occupies a clear, defensible white space in regenerative agriculture R&D — no existing institution conducts integrated, long-term research at the nexus of climate change, water sustainability, soil health, and agricultural economics in irrigated Western agriculture. This compendium synthesizes findings across eight critical topic areas to support interactive visualizations, investor-facing materials, and strategic positioning. The data reveals a convergence of crises in California's Central Valley — **$2 million acre-feet of annual groundwater overdraft, up to 900,000 acres of farmland facing retirement, and $320 billion in global climate-disaster losses in 2024** Munich Re **alone** — alongside a regenerative agriculture investment gap where current funding meets less than 10% of the estimated $200–450 billion needed annually for global agricultural transition. Fairr

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Topic 1: The invisible engine beneath our feet — soil microbiome science

A single gram of healthy soil contains over **1 billion bacterial and archaeal cells**, trillions of viruses, and tens of thousands of protists. Lawrence Livermore National … This staggering biodiversity drives the carbon cycle that makes agriculture possible — and that regenerative practices can dramatically enhance.

Mycorrhizal fungi as carbon highways

The landmark 2023 study by Hawkins et al. in *Current Biology* quantified for the first time that global plant communities allocate **13.12 gigatons of CO_2-equivalent per year** to mycorrhizal fungal networks — roughly **36% of current annual fossil fuel emissions**. Arbuscular mycorrhizal (AM) fungi alone receive 3.93 Gt, with ectomycorrhizal fungi receiving 9.07 Gt. PubMed Plants transfer **4–20% of their photosynthetically fixed carbon** to these fungal partners, Springer which extend several kilometers of hyphae through each gram of soil. Wiley Online Library The Society for Protection of Underground Networks (SPUN) calculated that global mycorrhizal mycelium in the top 10 cm of soil stretches approximately **450 quadrillion kilometers** — roughly half the width of the Milky Way galaxy. Spun

A 2024 study by Kakouridis et al. in *New Phytologist* demonstrated that AM fungi transferred 0.77 mg of carbon per gram of soil in just six weeks, with **33% deposited in mineral-associated pools** Wiley Online Library — the most stable form of soil carbon storage. The phylum Glomeromycota currently comprises roughly **338 described AMF species** that associate with approximately 80% of vascular plant families. Frontiers

The "Wood Wide Web" hypothesis remains under active scientific debate. Karst et al. (2023, *Nature Ecology & Evolution*) found that claims about common mycorrhizal networks facilitating cooperation between trees are "largely disconnected from evidence," Nature while Simard (2025, *Frontiers in Forests and Global Change*) defended the research, Frontiers and Merckx et al. (2024, *Nature Plants*) provided evidence through mycoheterotrophic plants that fungal networks do transfer meaningful carbon between organisms. Phys.org

Bacteria, biomass, and the necromass revolution

Microbial biomass carbon constitutes **0.5–5% of total soil organic carbon**, (ScienceDirect) with fungi contributing approximately 70% of that biomass by weight despite bacteria being far more numerous by cell count. (Springer) The fungi-to-bacteria ratio serves as a key soil health indicator: intensively managed cropland shows ratios below 1:1 (bacterial-dominated), while deciduous forests reach **5:1 to 10:1** and coniferous forests can reach 100:1 to 1000:1. (MicroBiometer) (Envirothonpa) Higher ratios correlate with greater carbon storage potential. (PubMed Central)

Perhaps the most striking recent finding is that **dead microbial biomass (necromass) constitutes up to 50% of all soil organic matter** — making it one of the largest organic carbon stocks on the planet. (Lawrence Livermore National …) This insight, from Lawrence Livermore National Laboratory research published in *Nature Reviews Microbiology*, fundamentally reframes how soil carbon accumulates: microbes don't just cycle carbon, their remains become the carbon bank.

Nitrogen-fixing bacteria in legume-Rhizobium symbioses deliver **100–300 kg of nitrogen per hectare per year** under good conditions, (Indogulf BioAg) with free-living fixers contributing an additional 20 kg N/ha/year. (Nature) About 80% of biologically fixed nitrogen entering soil ecosystems comes from these symbiotic relationships. (IntechOpen) (ScienceDirect)

Regenerative practices rebuild microbial communities on measurable timelines

A global meta-analysis of 60 studies found that conservation tillage increased overall microbial biomass by **37%** compared to conventional tillage, with fungal biomass rising 31% and bacterial biomass 11%. (ScienceDirect) Bacterial communities can recover within **1–3 years** of ceasing tillage, but AM fungal biomass requires significantly longer (ResearchGate) — a critical finding for transition planning. UC Davis long-term studies showed conservation agriculture increased soil carbon by approximately **45% over 8 years** in the San Joaquin Valley. (UC Davis)

Glomalin-related soil protein (GRSP), the glycoprotein produced by AM fungi, (ScienceDirect) persists in soil for **6–42 years** (ScienceDirect) and can account for **4–5% of soil carbon** in forest soils (Nature) and up to 19% in deep agricultural soils. (Nature) This recalcitrant protein creates hydrophobic coatings around soil aggregates, (Frontiers) directly linking fungal health to soil structure, water infiltration, and carbon storage. Forest soils contain glomalin concentrations **2.35–2.56 times higher** than farmland (PubMed Central) — a gap regenerative practices can begin closing.

Visualization-ready metrics

Metric	Value	Source
Bacteria per teaspoon of soil	Up to 1 billion	USDA NRCS
Microbial taxa per teaspoon	10,000–50,000	UC Davis
Global mycorrhizal carbon allocation	13.12 Gt CO_2e/year (36% of fossil fuel emissions)	Hawkins et al. 2023
Fungal biomass (global)	~12 Gt C vs. ~7 Gt C bacteria	Bar-On et al.
Microbial necromass share of SOM	Up to 50%	LLNL
Tillage → no-till microbial biomass gain	+37%	ORNL meta-analysis, 2020
Conservation ag → SOC gain (8 years)	~45% increase	UC Davis

Topic 2: California's Central Valley faces an existential water reckoning

The Central Valley produces **25% of America's food on just 1% of farmland**, Farmonaut® FOX Weather including 40% of all U.S. fruits and nuts. The Edison Institute CA This $50 billion agricultural economy now confronts a convergence of groundwater depletion, regulatory mandates, land subsidence, and climate change that threatens its fundamental viability.

SGMA enforcement is accelerating

The Sustainable Groundwater Management Act Allen Matkins Farmonaut® requires 21 critically overdrafted basins to achieve PPIC sustainability by **2040**. The southern San Joaquin Valley alone overdrafts groundwater by nearly **2 million acre-feet annually** Bay Nature — more than 10% of net water use. PPIC The Public Policy Institute of California (PPIC) estimates that ending this overdraft will require pumping cutbacks of approximately **2.5 million acre-feet per year**, PPIC with average annual farm water availability declining by **3.15 million acre-feet by 2040**. PPIC

Enforcement has moved from planning to action. The Tulare Lake Subbasin became the first basin placed on SGMA probation Northkingsgsa in **April 2024**, followed by the Tule Subbasin in September 2024. Valley Ag Voice The Kern County Subbasin — California's largest Valley Ag Voice at 2,834 square miles Kerngsp — began probationary hearings in February 2025. Valley Ag Voice Across these basins, non-exempt extractors began measuring groundwater use on January 1, 2025, Maven's Notebook with first reports due February 2026.

The fallowing crisis is measured in hundreds of thousands of acres

PPIC projects **500,000 to 1 million acres** of irrigated Central Valley farmland will need permanent retirement by 2040 (High Country News) — with the worst-case scenario reaching 900,000 acres, roughly 17.7% of the valley's agricultural fields. (Valley Ag Voice) In Kern, Kings, and Tulare counties alone, over **625,000 acres** of fallowing is expected. The economic consequences are severe: PPIC's worst case models **~50,000 jobs lost** and a 2.3% decline in regional economic activity. (PPIC)

The fallowing is already underway. Kern County removed nearly **17,000 acres** from production in 2024 alone. (Valley Ag Voice) Tulare County lost 12,485 acres to water-related removal that year. (Recorder Online) The shift from annual to permanent crops has "hardened" water demand — almond orchards cost $9,000–$12,000 per acre to plant and $3,000–$4,000 per acre to remove, making flexible fallowing of tree crops economically devastating. Kern County's bearing almond acreage fell 13,000 acres in a single year, (Valley Ag Voice) and its wine grape value collapsed from **$132 million in 2023 to $27 million in 2024** (Hanford Sentinel) — an 80% drop. (Agri-Pulse)

Land values in overdrafted basins have cratered, falling **up to 75%** depending on commodity and water source. Properties solely reliant on pumped groundwater experienced 50%+ value declines. (Agri-Pulse) Pistachio orchards in Kern dropped 15.5–47.6% in just six months. (SJV Water)

Tulare Lake's return is a warning from the past

In early 2023, atmospheric rivers and record Sierra snowpack refilled Tulare Lake — historically the largest freshwater lake west of the Mississippi (Smithsonian Magazine) at 512,000 acres — for the first time in over 25 years. The lake expanded to cover approximately **111,000 acres**, (Smithsonian Magazine) inundating ~94,000 acres of farmland (Travelnytimes) with estimated agricultural losses of **$900 million to $2 billion**. Kings County crop receipts dropped 17% to $2.15 billion.

Land subsidence from decades of groundwater overdraft had lowered basin elevations by **10–15 feet** in some areas, (CNN) worsening flood risk. The recurrence interval for lakebed flooding is roughly every 15 years, (California WaterBlog) making this a predictable consequence of the collision between overextracted aquifers and climate-amplified precipitation.

Infrastructure is literally sinking

Since the 1920s, excessive pumping has caused land to sink **up to 28 feet** in sections of the San Joaquin Valley. (NASA Earth Observatory) A landmark Stanford study published in *Nature Communications Earth & Environment* (November 2024) found that **14 km³ of total subsidence occurred between 2006 and 2022** — equaling the volume measured during 24 years of historic monitoring. (Nature) Peak rates near Corcoran and El Nido have reached **up to 2 feet per year**. (NASA Earth Observatory)

The Friant-Kern Canal, which serves over 1 million acres and 250,000 residents, has **lost more than 60% of its original capacity** due to subsidence, (Friantwater) dropping from 4,000 cfs to roughly 1,600 cfs. (Water Education Foundation) (Manteca Bulletin) Phase 1 repairs covering just 10 miles cost **$325 million** — 74% over the original estimate — and surrounding areas have already sunk another foot, crimping the rebuilt canal's capacity by 250 cfs (SJV Water) before full use. The California Aqueduct has lost **20% of its capacity** near Avenal. (Citrus Heights Messenger) Total repair costs across the valley's conveyance infrastructure run into the billions, with the state committing over $200 million in recent budgets and still falling short.

Topic 3: How leading institutions communicate complex data — and what Edison can learn

The most effective climate-agriculture organizations share a common playbook: iconic signature visualizations, layered depth for multiple audiences, open data commitments, and professional design partnerships. The Edison Institute can learn from — and improve upon — these models.

The power of a single iconic image

The Stockholm Resilience Centre's **planetary boundaries diagram** is arguably the most influential data visualization in environmental science. Created in partnership with Azote design studio and Globaïa, the circular radar diagram with color-coded wedges (green/yellow/red) communicates the entire state of Earth's systems at a glance. Globaia Updated through multiple versions (2009, 2015, 2023, 2025), it now shows **7 of 9 boundaries transgressed**. Globaia The lesson: invest in one definitive visualization that encapsulates the research narrative, then update it regularly. Edison should develop an equivalent "Central Valley Systems Health" diagram.

Data tools that drive decisions

The World Resources Institute (WRI) operates the gold standard in interactive environmental data platforms. Atlas Their **Aqueduct** tool maps water risk at sub-basin level globally and has been cited by The Economist, Bloomberg, and the New York Times. World Resources Institute **Resource Watch** integrates 300+ datasets with Mapbox-powered interactive mapping. World Resources Institute All data is freely available under Creative Commons — a radical openness that maximizes reach and credibility. World Resources Institute WRI's multi-platform strategy (different tools for different questions) demonstrates that purpose-built tools outperform one-size-fits-all dashboards. Atlas

Our World in Data achieves similar impact through a different approach: **custom-built "Grapher" charts** that are interactive, embeddable, and each have a unique URL. Our World in Data Their long-form articles combine narrative text with multiple interactive charts and transparent source citations, creating comprehensive topic pages that are regularly updated. Every chart is CC-BY licensed Our World in Data with a fully documented data pipeline on GitHub. Our World in Data

Outcomes-based verification creates market value

The Savory Institute's **Ecological Outcome Verification (EOV)** protocol Gitbook measures 24+ data points across soil health, biodiversity, and water cycle function Ata — then trends them over time to measure improvement. This outcomes-based (rather than practice-based) approach Landtomarket connects directly to their Land to Market program, creating a data pipeline that serves both research and brand storytelling. The methodology is published transparently in a public GitBook.

Rodale's 40-year lesson: time is the most valuable dataset

The Rodale Institute's Farming Systems Trial, running since 1981, demonstrates that **long-term data is the ultimate differentiator**. Even simple bar charts become compelling when backed by four decades of comparison. Their communication strategy uses a "headline finding → supporting chart → plain-language explanation" format, with downloadable PDF reports as the primary funder-facing vehicle.

Strategic recommendations for Edison Institute communications

- **Develop one iconic visualization** that encapsulates the Central Valley climate-water-soil-economics nexus, created with a professional design studio
- **Build purpose-specific interactive tools** (Flourish for scrollytelling, Mapbox for spatial data, D3.js for custom charts) rather than one generic dashboard
- **Use scrollytelling** for key research narratives — tools like Flourish, Shorthand, or Vev make this accessible without engineering teams (Flourish)
- **Publish data openly** from day one — this signals institutional quality and accelerates adoption
- **Create polished annual impact reports** in PDF format with designed infographics for investment committee meetings
- **Layer communication for audiences**: hero metrics and trend lines for investors; raw data access for researchers; shareable single-chart stories for media; site-specific tools for farmers
- **Start collecting longitudinal data immediately** — Rodale's 40-year dataset is irreplaceable. Every month of data collection compounds in value

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Topic 4: Regenerative agriculture delivers hard ROI — with important caveats

The economic case for regenerative agriculture is strengthening, with multiple institutional studies documenting **$44–$65 per acre increases in net income** for adopters. However, the evidence base remains uneven, transition costs are real, and the most-cited profitability figures include business model changes (stacked enterprises, direct marketing) alongside agronomic improvements.

The Soil Health Institute's 100-farm study is the strongest evidence

The Soil Health Institute's study of 100 farms across 194,003 acres in 9 Midwest states — funded by Cargill — found average net income increases of **$51.60/acre for corn and $44.89/acre for soybeans**. (AGDAILY) Some 85% of corn farmers and 88% of soybean farmers saw increased net income. (Soilhealthinstitute) Farmers averaged 19 years of no-till and 9 years of cover crops, (AGDAILY) suggesting these are mature-adoption outcomes. A follow-up 30-farm study across 20 states found an average **$65/acre increase** across diverse crops. (Soilhealthinstitute)

BCG and WBCSD's 2023 report "Cultivating Farmer Prosperity" estimated transition costs of up to **$40/acre** with a 3–5 year payback period, and found long-term profit increases of up to **120% above conventional** (Veriswp) for Kansas wheat farmers. (As You Sow +2) McKinsey's 2024 analysis of US corn and soy projected **$20–$60/acre annual returns** with 10–30% yield improvements, estimating **$250 billion in aggregate economic value** over a decade at 80% adoption. (Agtechnavigator)

The peer-reviewed LaCanne & Lundgren study (2018, *PeerJ*) of 20 farms found regenerative fields had **29% lower grain production but 78% higher profits** Acresusa PeerJ — a figure frequently cited but requiring context: profitability included premium marketing revenue from stacked enterprises, not just agronomic gains. Acresusa EcoWatch

Input cost savings are well-documented

Fertilizer reductions of **41–53%** across major crops (corn, soybeans, wheat, cotton) are consistently reported. Ecosystems United A Missouri farmer with 1,400 acres documented **$30/acre savings** ($42,000/year) after 10 years of cover crops. KBIA Herbicide use reductions of 50% are reported by established adopters. Cover crop costs run **$16–$40/acre** (seed through termination), with break-even typically reached at approximately 3 years. Ecosystems United

Gabe Brown's ranch in North Dakota — perhaps the most famous regenerative farm globally — increased soil organic matter from **1.9% to 6.1%** and water infiltration from 0.5 to 8 inches per hour over 25+ years. California State University, Chico However, specific revenue-per-acre figures are not publicly disclosed, and claims of profitability have not been independently audited.

Carbon credit revenue remains modest but growing

Indigo Agriculture, the largest issuer of registry-verified agricultural soil carbon credits, has enrolled **8+ million acres across 28 states** and issued over 2 million metric tons of verified CO_2e credits iGrow News through 2026. Farmer payment rates reached **$30 per credit** in 2022, with 75%+ revenue share. Indigo Ag At typical sequestration rates, this translates to roughly **$4.50–$11.25/acre/year** — meaningful but below the $35/acre threshold most farmers say they need to participate. AgFunderNews Total farmer payments through 2023 reached $12+ million collectively. Carbon Credits Major buyers include Microsoft (12-year agreement for 2.85 million credits), Grand View Research iGrow News JPMorgan Chase, and Barclays. CNBC

Crop insurance remains a structural barrier

No actuarial premium discount currently exists for regenerative practices in the Federal Crop Insurance Program. The system actively penalizes some regenerative practices: intercropping can disqualify claims, and late cover crop termination risks coverage denial. Civil Eats The Pandemic Cover Crop Program offered a temporary **$5/acre discount** that enrolled 12 million acres before expiring in 2023. U.S. Congressman Sean Casten UC Berkeley researchers with $2.2M in USDA/FFAR funding are working to quantify risk reduction for actuarial models. Berkeley Meanwhile, drought indemnities rose **400%** from $325.6 million (1995) to $1.65 billion (2020), Landstewardshipproject and FCIP costs are projected to increase 37% without climate adaptation practices. U.S. Congressman Sean Casten

Summary of key financial metrics

Metric	Value	Source
Net income increase (corn)	$51.60/acre avg	SHI 100-farm study, 2021
Net income increase (soybeans)	$44.89/acre avg	SHI 100-farm study, 2021
Net income increase (diverse)	$65/acre avg	SHI 30-farm study, 2023
Transition cost	$40–$200/acre	BCG 2023; McKinsey 2024
Payback period	3–5 years	BCG; USDA-SARE
Long-term profit increase	Up to 120%	BCG/WBCSD 2023
Carbon credit revenue	~$4.50–$11.25/acre/year	Calculated from Indigo data
Aggregate US potential	$250B over decade	McKinsey 2024

Topic 5: CRARS and The Carbon Underground bring complementary strengths to Edison

The Center for Regenerative Agriculture and Resilient Systems at CSU Chico (California State University, Chico) is the **nation's first comprehensive university-based regenerative agriculture program**, (Chico State Today) while The Carbon Underground brings decades of strategic relationships across business, science, and finance. Together, they provide Edison with both scientific infrastructure and institutional credibility.

CRARS has built substantial infrastructure since 2019

Founded by Dr. Cynthia Daley (California State University, Chico) — a fourth-generation farmer (Chico State Today) who received CSU's inaugural **$100,000 Gayle E. Hutchinson Award for Excellence** (Chico State Today) — CRARS has secured **more than $30 million in grant funding** and advanced **119 projects across 34,000 acres in 16 California counties**. (Chico State Today) Key federal partnerships include a $7 million USDA NRCS Regional Conservation Partnership Program grant (California State University, Chico) and a $5 million USDA Partnership for Climate-Smart Commodities award. (California State University, Chico)

The center's **Soil Carbon Accrual Project (SCAP)** (Noregretsinitiative) uses flux tower (eddy covariance) technology across three major California growing regions — Sacramento Valley, San Joaquin Valley, and Palo Verde Valley — to continuously measure CO_2 flow, representing a significant advancement over conventional point-in-time soil carbon measurements. First towers were installed at the Blythe test site in November 2022.

CRARS maintains approximately **20+ mentor farmer demonstration sites** across California, including almond (NicNuts), rice (Massa Organics), dairy (Alexandre Family Farm), vineyard (Tablas Creek), and rangeland operations. Since 2023, the center partnered with a local bureau to award **50 Healthy Soils grants worth up to $100,000 each**, covering 4,550 acres. (Chico State Today) Dr. David Johnson, an adjunct professor, (California State University) brings the Johnson-Su Bioreactor composting system — producing fungal-dominant compost (Soil Centric) now used globally across multiple continents. (Rodale Institute)

The Carbon Underground connects science to capital and markets

Co-founded in 2013 Thecarbonunderground by Larry Kopald The Edison Institute (former advertising executive for McDonald's, Nike, and Honda, and board member of Oceana) and Tom Newmark (former CEO of New Chapter, chairman of Greenpeace Fund USA), The Carbon Underground operates as an outcomes-based nonprofit connecting business, science, finance, and government to soil restoration. Thecarbonunderground Their strategic alliances span Savory Institute, Rodale Institute, Kiss the Ground, and Regeneration International. Thecarbonunderground In 2017, TCU and CRARS co-created the **Regenerative Agriculture Definition** signed by 150+ companies, organizations, and scientists.

Edison Institute's current profile

The Edison Institute for Food & Environmental Security operates on **325 donated acres** in the Central Valley theedisoninstitute — described as "ground zero for the intersection of agriculture and climate change." The Edison Institute The Schuman Family Foundation contributed **more than $7 million in land and operating funds**. The Edison Institute Current research projects include citrus resilience trials on 100+ acres of conventionally grown oranges, agave as an alternative crop for an increasingly arid Central Valley, cover crop persistence research, and agricultural waste repurposing. The institute emphasizes four differentiators: independent research center status, full management control on dedicated acreage, a whole-systems approach to R&D, and commitment to long-term research measured in years rather than seasons. theedisoninstitute The Edison Institute

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Topic 6: A massive funding gap creates opportunity for well-positioned R&D centers

The regenerative agriculture investment landscape is growing rapidly but remains dramatically undersized relative to need. FAIRR Initiative and PRI estimate the global transition requires **$200–$450 billion per year** for the next decade, yet current funding flows meet **less than 10% of that target**. This gap represents both urgency and opportunity for institutions that can demonstrate credible, measurable impact.

Private capital is accelerating from a small base

BCG's 2025 report identified a **$310 billion investment opportunity** in landscape-level regenerative transition. BCG The Regenerative Food Systems Investment Forum tracked **$1.17 billion** in tools and solutions deals in Q1 2025 alone, plus over $650 million in fund raises for regenerative and adjacent funds. AgFunderNews Key active funds include SLM Partners (**$760 million AUM**, SLM Partners 40% year-on-year growth in 2024), Conservationfinancenetwork Iroquois Valley Farmland REIT (**$124.4 million portfolio** across 36,150 acres Iroquois Valley delivering 9% annualized returns since 2007), Mad Capital's Perennial Fund II (**$50 million** backed by Rockefeller Foundation, Builders Vision, and Schmidt Family Foundation), Rockefeller Foundation and the AXA/Unilever/Tikehau fund seeded at **€300 million**. Veriswp A dedicated tracker identifies **24+ funds with $4.4 billion in combined AUM** specifically focused on building soil. Investinginregenerativeagriculture

Government funding reached historic levels — then hit uncertainty

The Inflation Reduction Act directed approximately **$19.5 billion** to climate-smart agriculture through NRCS programs: USDA $8.45 billion for EQIP, $4.95 billion for RCPP, USDA $3.25 billion for CSP, and $1.4 billion for ACEP. In FY2024, EQIP approval rates reached **43–44%** (highest since FY2018) and CSP reached **53–55%** (highest since FY2012). In December 2025, NRCS announced a **$700 million Regenerative Pilot Program** ($400M EQIP, $300M CSP) requiring whole-farm assessment and soil health testing. However, reports indicate IRA funds were **frozen as of May 2025**, creating significant uncertainty.

The USDA's Climate-Smart Commodities program deployed **$3.1 billion across 141 projects** KBIA covering 25+ million acres. Conservationfinancenetwork The Foundation for Food and Agriculture Research The Land Institute leverages **$1.40 for every federal dollar** invested. The Breakthrough Institute Agricultural R&D delivers extraordinary returns — a Cambridge study estimated benefit-cost ratios of **33:1** Cambridge Core for Global South investments.

Carbon markets are volatile but trending toward growth

The voluntary agriculture carbon credit market reached **$36.1 million in 2024** and is projected to grow to **$648.3 million by 2034** at 31.9% CAGR. Global Market Insights Overall voluntary carbon market average prices declined 20% to **$4.80/ton** in 2024, Carbon Credits though farm-level payment programs still pay **$10–$35/ton**. The broader AFOLU market saw over **$8 billion in transactions** in 2024, projected to reach $140 billion by 2035. Bis Research

Philanthropic capital is concentrating in regenerative agriculture

The Bezos Earth Fund committed **$1 billion** to its Future of Food program. Rockefeller Foundation committed $105 million to healthy and sustainable food. The Agroecology Fund Agroecology Fund nearly doubled its all-time grantmaking in 2022–2024 after receiving $10 million from Waverley Street Foundation (Laurene Powell Jobs) and $9 million from Ballmer Group. Funders for Regenerative Agriculture (FORA) grew to **60+ members** by 2024. Yet the Global Alliance for the Future of Food reports that only **3% of climate finance** goes to food systems transformation globally. Inside Philanthropy

Language that resonates with different funder types

Impact investors respond to risk-adjusted returns, farmland as inflation hedge (5–6% annual appreciation plus cash flow), Conservationfinancenetwork and portfolio diversification. Philanthropic funders prioritize systems-level impact, racial equity, and catalytic capital that de-risks private investment. Government funders need carbon sequestration quantification (metric tons CO_2e), farmer/acre enrollment numbers, cost-benefit ratios, and Farm Bill alignment. The most effective R&D institutes use **blended finance structures** combining concessional capital PubMed Central (foundations, government first-loss pools) with market-rate institutional investment.

Topic 7: Edison occupies clear white space in the competitive landscape

A systematic analysis of the top regenerative agriculture research institutions reveals a critical gap: **no existing organization conducts integrated, long-term research studying climate change, soil health, water systems, and agricultural economics together in irrigated Western agriculture.** Every major competitor focuses on one or two dimensions, none addresses California's Central Valley, and nearly all concentrate on dryland grain or grazing systems irrelevant to the state's $50+ billion specialty crop economy.

The competitive landscape clusters in the East and Midwest

The Noble Research Institute (Oklahoma, ~$60M annual budget, 13,500 acres) focuses exclusively on regenerative ranching — important but limited to grazing lands with no irrigated agriculture engagement. The Soil Health Institute (North Carolina) has built the most comprehensive soil measurement database in North America across 124 long-term research sites GuideStar but focuses narrowly on measurement science without integrating water, climate, or economic modeling. Rodale Institute (Pennsylvania, 44+ years of FST data) concentrates on organic grain systems in the Eastern US; Mad Agriculture their California Organic Center is in coastal Ventura County, far from the Central Valley.

The Land Institute The Land Institute (Kansas) conducts deep perennial grain breeding research The Land Institute Wikipedia — an important agricultural moonshot — but is narrowly focused on new crop development rather than existing farming systems. The Savory Institute (Colorado) operates a global hub network Savory Institute for holistic grassland management but lacks original research infrastructure, peer-reviewed trials, or engagement with irrigated agriculture. Understanding Ag/Soil Health Academy provides Soil Health Academy respected practitioner-led education Understanding Ag but generates no original data. Kiss the Ground and the Regenerative Organic Alliance Pocono Organics serve media/certification functions without research capacity.

California has research assets but no integrated Central Valley center

UC Davis offers broad agricultural sustainability research across departments but is a university with fragmented faculty-driven programs, not a dedicated R&D center. UC Merced, newly designated as an Agricultural Experiment Station campus, is building regenerative agriculture capacity but remains in very early stages. UC Agriculture and Natural Re… CRARS at CSU Chico has the strongest California-specific regenerative research portfolio but focuses on Northern California. California State University, Chico

The USDA's Long-Term Agroecological Research (LTAR) Network spans 19 sites nationally, LTAR Network but Western representation is limited to rangeland sites in New Mexico and Idaho — **none specifically studies the climate-agriculture-water nexus in irrigated California systems**. The Columbia Basin Agricultural Research Center in Oregon (since 1931) focuses on dryland cereals. There is no long-term integrated research facility anywhere in the Central Valley.

Six areas of white space for Edison

Integrated whole-systems research. No institution conducts simultaneous, place-based field research on climate change × soil health × water systems × farm economics in irrigated Western agriculture.

Irrigated specialty crop focus. Nearly all competitors study dryland grain, grass, or ranching. California's 250+ specialty crops (almonds, grapes, tomatoes, pistachios, citrus) represent billions in value with no dedicated regenerative research infrastructure.

SGMA/groundwater sustainability integration. No regenerative agriculture institution researches how soil health practices interact with groundwater recharge, managed aquifer recharge, or SGMA compliance — the most pressing agricultural policy challenge in California.

Central Valley as global natural laboratory. The Mediterranean climate represents the climate future for many global agricultural regions. The extreme water stress, crop diversity (400+ products), and socioeconomic complexity create research conditions replicable nowhere else.

Technology-enabled regenerative research. A gap exists between the "low-tech/nature-based" regenerative movement and precision agriculture (remote sensing, AI-driven water management, soil sensors). Edison can bridge this divide.

Economic transition modeling. No institution systematically models the economics of transitioning irrigated specialty crop operations to regenerative practices under declining water availability — an urgent question for Central Valley growers.

Positioning statement

The Edison Institute can credibly claim to be **"the first dedicated R&D center conducting integrated, long-term whole-systems research at the nexus of climate change, water sustainability, soil health, and agricultural economics in irrigated Western agriculture — located in the heart of America's most productive and most water-stressed farming region."** No existing institution combines Central Valley location, irrigated specialty crop focus, integrated climate-water-soil-economics research, long-term field trial infrastructure, and independence from a single university system.

Topic 8: Climate disruptions are reshaping agricultural supply chains with billions in losses

Climate-driven agricultural supply chain disruptions have escalated from episodic shocks to systemic risks, with global losses reaching **$320 billion in 2024** (Munich Re) Munich Re and FAO estimating **$3.8 trillion in cumulative agricultural disaster losses** from 1991–2021 ($123 billion/year average). These events demonstrate the fragility of conventional agricultural systems and the economic imperative for resilience.

A cascade of crises from 2021 to 2025

Winter Storm Uri (February 2021) caused **over $608 million** in Texas agricultural losses — $230 million in citrus (Rio Grande Valley lost virtually 100% of Valencia oranges), $228 million in livestock, $150 million in vegetables — while shutting down 75% of U.S. chemical production capacity. The 2022 European heat wave, described as the continent's worst drought in 500 years, slashed EU corn production by **16%**, with Italy's corn yields dropping 45% and France's falling 18.5%. Cumulative European drought production losses from 1983–2009 exceeded $14 billion across major producing countries.

The 2022 Western U.S. drought drove California rice planting down by **50%**, causing $703 million in lost economic activity and 5,300 rice-related job losses. Texas cotton saw record **69% crop abandonment**. The USDA's smallest hard red winter wheat crop since 1963 combined with drought-driven supply disruptions to push food CPI up **10.4%** in 2022, with grocery prices rising 12.2%.

The Panama Canal drought (2023–2024) reduced daily transits by **36%** (from 36–38 ships to 22–24), adding approximately **$10/ton to grain shipping costs** and forcing rerouting around the Cape of Good Hope at an additional **$30,000 per day per vessel**. Auction slots for canal passage sold for $1–2 million each — untenable for thin-margin agricultural trade. India's July 2023 rice export ban — affecting 40% of global rice trade — pushed global rice prices to **15-year highs** with 12–32% price spikes, costing consumers an estimated $315 million annually.

The West African cocoa crisis drove prices from **$2,500 to $12,931 per metric ton** in 2024 — a tripling driven by extreme rainfall, drought, and an unprecedented 50°C heatwave. Mediterranean drought pushed EU olive oil prices up **50%** in a single year. Japanese rice prices rose 48%; South Korean cabbage surged 70%.

Insurance payouts are on an unsustainable trajectory

U.S. crop insurance indemnity payments hit a record **$19.1 billion in 2022**, up from approximately $8 billion in 2019. Over 2021–2023, crop insurance paid **$44.4 billion** — four-fifths of total farm program payments. The 20-year trend shows roughly 20% annual growth in indemnities. Munich Re's global natural catastrophe data shows insured losses exceeding $100 billion annually is becoming standard, with 2024's $140 billion the third-costliest year since 1980.

Research by Kotz et al. (2025, *Environmental Research Letters*) analyzed 16 examples of food price rises tied to extreme weather between 2022 and 2024, finding that high temperatures increased food inflation "persistently" for 12 months after extremes in both high- and low-income countries. McKinsey's agriculture practice found that **40%+ of variation in crop yields** in recent decades was caused by precipitation and temperature extremes.

The resilience case for regenerative agriculture

These disruptions make a powerful case for the Edison Institute's research agenda. Regenerative practices demonstrably improve drought resilience (Rodale's FST shows organic outperforming conventional in drought years), reduce flood damage through improved infiltration (Gabe Brown's ranch: 0.5 → 8 inches/hour), and lower input cost exposure to fertilizer price spikes (41–53% reduction in synthetic fertilizer use). With USDA crop insurance costs projected to increase **37% without climate adaptation practices**, the value of research quantifying regenerative resilience — in the specific context of California's irrigated specialty crop systems — has never been higher.

Conclusion: A convergence of crises creates a convergence of opportunity

This compendium reveals five interconnected realities that define the Edison Institute's strategic context. First, **the science is clear but incomplete** — we know mycorrhizal networks move 13 Gt of CO_2-equivalent annually and that conservation tillage increases microbial biomass by 37%, but we lack integrated field data on how these processes interact with water systems and farm economics in irrigated Western agriculture. Second, **the Central Valley crisis is accelerating faster than solutions** — 2 million acre-feet of annual overdraft, up to 900,000 acres facing retirement, and billions in subsidence-related infrastructure damage create existential urgency. Third, **the economic case for regenerative transition is strong but underdeveloped for California's specialty crop systems** — $44–$65/acre income gains documented in Midwest grain systems need equivalent evidence for almonds, grapes, citrus, and vegetables under water constraints. Fourth, **the investment landscape is capital-rich but evidence-starved** — a $200–450 billion annual funding need versus less than 10% current flows, with investors specifically seeking the kind of integrated, place-based, long-term data that Edison can uniquely generate. Fifth, **no competitor occupies Edison's white space** — the convergence of Central Valley location, irrigated specialty crop focus, whole-systems integration, and long-term research commitment is genuinely unprecedented.

The path forward is to begin generating longitudinal data immediately. As Rodale's 40-year trial demonstrates, time is the most valuable dimension in agricultural research. Every season of integrated measurement across climate, soil, water, and economics compounds Edison's strategic advantage and produces the evidence base that funders, policymakers, and farmers urgently need.